

A multi-source information transfer learning method with subdomain adaptation for cross-domain fault diagnosis

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ABSTRACT

In modern industrial equipment maintenance, transfer learning is a promising tool that has been widely utilized to solve the problem of the insufficient generalization ability of diagnostic models, caused by changes in working conditions. However, owing to the single knowledge transfer source and fuzzy marginal distribution matching, the ability of traditional transfer learning methods for cross-domain fault diagnosis is not ideal. In practice, collecting multi-source data from different scenarios can provide richer generalization knowledge, and fine-grained information matching of relevant subdomains can achieve more accurate knowledge transfer, which is conducive to the improvement of the cross-domain fault diagnosis performance. To this end, a multi-source subdomain adaptation transfer learning method is proposed to transfer diagnostic knowledge from multiple sources for cross-domain fault diagnosis. This approach exploits a multi-branch network structure to match the feature spatial distributions of each source and target domain separately, where the local maximum mean discrepancy is used for fine-grained local alignment of subdomain distributions within the same category of different domains. Moreover, the weighted score of a source-specific is obtained according to its distribution distance, and multiple source classifiers are combined with the corresponding weighted scores for the joint diagnosis of the device status. Extensive experiments are conducted on three rotating machinery datasets to verify the effectiveness of the proposed model for cross-domain fault diagnosis.

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1. Introduction

Rotating machinery is essential mechanical equipment in industrial applications, and its safe and reliable operation guarantees high-quality industrial products. Therefore, real-time condition monitoring and fault diagnosis of mechanical equipment are critical. With the digitalization and intellectualization of industrial manufacturing, fault diagnosis and health management of equipment have shifted from model-based and signal analysis methods to data-driven solutions [1]. Deep learning has already been proven to be an efficient data-driven technology for fault detection, owing to its remarkable advantages, such as automatic feature extraction and end-to-end process architecture [2]. When large-scale labeled data for supervised learning are available, the fault diagnosis task can be effectively solved. Han et al. [3] proposed an improved threshold adaptation deep belief network (DBN), where multiscale permutation entropy (MSPE)

and wavelet packet energy entropy (WPEE) were extracted from vibration signals as network input features for gearbox state recognition. Shi et al. [4] utilized compression sensing technology to reduce the data dimension combined with WPEE extraction features to train the autoencoder network and effectively diagnose the fault of the rotating bearing. Zhang et al. [5] developed a convolutional neural network (CNN) with a wide filter for the fault identification of rotating machinery with good anti-noise ability. For the above diagnostic tasks, which can obtain training data under the same conditions as the test environment, the diagnosis accuracy reached a satisfactory level.

However, in the actual industry, owing to the interference of environmental noise and frequent changes in equipment operating conditions, the collected data for training and testing have a certain degree of domain shift, which causes the diagnostic performance of the model learned from the training dataset to suffer significant degradation when applied to the test scenario [6,7]. Because the distribution obeyed by the training and testing data is inconsistent, learning a discriminative model is known as domain adaptation, which learns to project differently distributed data to a common high-dimensional space, to reduce domain

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shift, and the generalization ability of the model in different test scenarios can be enhanced [8]. Traditional deep models that do not consider domain shifts are unsuitable for real scenarios. Unsupervised domain adaptation (UDA), a popular transfer learning strategy, provides a promising direction for more practical fault diagnosis applications. Two mainstream UDA methods have been widely reported in literature. The first is based on difference measurement; cross-domain differences are reduced by embedding distance measurement algorithms such as KL divergence [9], maximum mean discrepancy (MMD) [10–12], and central moment difference (CMD) [13] into the adaptation layer of the deep model. The other is based on adversarial training, which is trained by the game between the feature generator and the domain recognizer so that the learned features are domain invariant. In recent years, researchers have explored the introduction of UDA based on distance measurements and adversarial training for cross-domain diagnosis tasks [14,15]. Wen et al. [16] developed a sparse autoencoder network based on the MMD metric, where the extracted domain-invariant features were used for fault monitoring of bearing components. Zhu et al. [17] introduced a multikernel MMD into a CNN to generalize the diagnosis ability of the model in different working states of rolling bearings. Han et al. [18] introduced an adversarial learning strategy into a deep CNN for the intelligent damage recognition of a gearbox. Guo et al. [19] used adversarial training to learn domain-invariant representation and applied MMD to further improve domain adaptation performance to combine two formal domain adaptation methods for fault diagnosis under variable working conditions. In [20], joint distribution domain adaptation was extended to deep models for fault identification of rotating bearings.

Although scholars have made some achievements in fault diagnosis based on UDA, they only considered the knowledge transfer of a single source domain, which usually cannot provide sufficient relevant knowledge for the target diagnosis task, resulting in unsatisfactory diagnosis results. In practice, multiple source domains with different underlying distributions can be collected from various machine-operating scenarios to compensate for the lack of diagnostic knowledge of a single source domain. Mansour et al. [21] considered that an ideal target distribution can be approximately expressed by a weighted combination of multiple source distributions, which provides theoretical support for multi-source transfer learning. Some studies have proven that multi-source UDA (MUDA) is valuable in image recognition and text analysis for improving model performance [22–24]. Xu et al. [25] proposed the use of adversarial training on each pair of source and target domains, and thereafter combined the multi-source classifier with the domain perplexity score to obtain the target label. Lu et al. [26] explored the combination of multi-source domain knowledge and fuzzy transfer learning problems. However, it is difficult to narrow the distribution differences among all domains. There are few reports on fault diagnosis based on MUDA, which is challenging. As shown intuitively in Fig. 1, it is usually impossible to completely eliminate the shift between a single source and target domain. A greater degree of mismatch may occur when trying to align multiple source and target domains, which leads to model performance degradation. It is easier to extract the domain-invariant representations of each source and target domain than to extract the common domain-invariant representations of all domains [27]. In addition, the previous UDA method mainly aligns the marginal distribution from the perspective of global domain adaptation, ignoring the distribution relationship of relevant subdomains of the same category in different domains, which may confuse the diagnostic knowledge of different classes and lead to negative migration. Both local and global distributions can be aligned

by matching the relevant subdomains of different category features [28]. Therefore, fine-grained information transfer between related subdomains should be considered to improve the generalization performance of the cross-domain diagnostic model.

Based on the above investigation, two shortcomings exist in the traditional UDA diagnosis method: (1) insufficient transfer knowledge of the single source domain and (2) fuzzy matching of the global marginal distribution. To compensate for the aforementioned two weaknesses and more effectively transfer diagnostic knowledge that can be identified to the target domain from multiple source domains, a novel multi-source subdomain adaptation (MSSA) transfer learning method is proposed in this study, for the condition recognition of rotating machinery. The approach applies multiple branch structures to narrow the distribution difference between each pair of source and target domains in parallel so that domain-invariant features can be extracted from multiple source distribution data. In addition, local MMD (LMMD) is introduced to the fine-grained local alignment of subdomain distributions within the same category of different domains to achieve more accurate status knowledge transfer. Finally, the weighted score of a source-specific is obtained according to the distribution distance between the source and the target domain, and multiple source classifiers are combined with the corresponding weighted scores for the joint diagnosis of the device status. The contributions of this study are as follows:

(1) An ensemble network model with a shared subnet and multiple source-specific subnets is constructed to map each pair of source and target domains to a specific feature space respectively to eliminate domain shift, which effectively alleviates the difficulty of aligning multiple domain distributions simultaneously.

(2) An improved distribution distance measurement algorithm, LMMD, is applied for fine-grained matching of the distribution of relevant subdomains from the perspective of local domain adaptation, which enables the accurate transfer of mechanical state information.

(3) Based on the principle that the combination of multi-source distributions can approximate the target distribution, the multiple source-specific classifiers are combined with the corresponding weights obtained according to the distribution distance of each pair of source and target domains to perform a joint diagnosis of the mechanical equipment.

The remainder of this paper is organized as follows. The problem setup and related basic theories are introduced in Section 2. The network architecture and optimization objective of the proposed method are described in detail in Section 3. In Section 4, the experimental verification and comparative analyses are presented. Finally, the conclusion is presented in Section 5.

2. Preliminaries

2.1. Problem setup

In the fault diagnosis application of MUDA, multiple training data sources with ground-truth labels and unlabeled testing data source can be collected from different equipment operating conditions. Specifically, N source domain data $\{D_{sj}\}_{j=1}^N$ are sampled from relevant but different distributions $\{P_{sj}(x, y)\}_{j=1}^N$, where $D_{sj} = \{x_i^{sj}, y_i^{sj}\}_{i=1}^{N_j}$ represents the j th source domain dataset, N_j represents the number of samples in j th source domain, $y_i^{sj} \in \{1, 2, \dots, k\}$ indicates the corresponding health status label, k represents the number of health status classes; The target domain dataset $D_t = \{x_i^t\}_{i=1}^{N_t}$ is sampled from the target distribution $P_t(x, y)$, where N_t represents the number of target domain samples, $P_{sj}(x, y) \neq P_t(x, y)$. Although there is no corresponding status

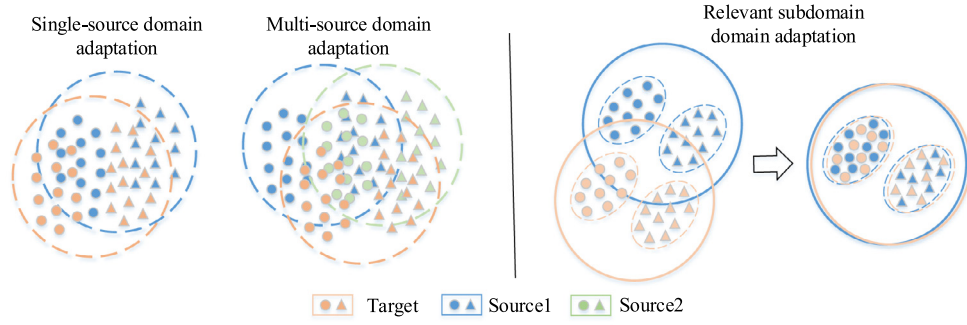


Fig. 1. Domain adaptation diagram.

label in the target domain, the target domain shares the status label space with all source domains, which is called closed-set domain adaptation. The purpose of cross-domain fault diagnosis is to construct an intelligent network model $\mathbf{y} = f(\mathbf{x})$ that can learn the domain-invariance and distinguishability characteristics, by reducing the difference between different distributions to minimize the risk of the target task $\mathbf{E}_{(\mathbf{x}, \mathbf{y}) \sim D_t} [f(\mathbf{x}) \neq \mathbf{y}]$ with multiple source supervision.

2.2. LMMD

In fault diagnosis based on domain adaptation, MMD, a non-parametric distance estimation method, has been universally applied to measure the distribution distance of different domains. At present, most MMD-based domain adaptation methods only perform global distribution alignment. This can be defined as follows.

$$D_{\mathcal{H}}(P_s, P_t) \triangleq \|\mathbf{E}_{\mathbf{x} \sim P_s} [\phi(\mathbf{x}^s)] - \mathbf{E}_{\mathbf{x} \sim P_t} [\phi(\mathbf{x}^t)]\|_{\mathcal{H}}^2 \quad (1)$$

where $\mathcal{H}$ indicates the reproducing kernel Hilbert space (RKHS) embedded with a characteristic kernel $\mathbf{k}$, and $\mathbf{k}(\mathbf{x}^s, \mathbf{x}^t) = \langle \phi(\mathbf{x}^s), \phi(\mathbf{x}^t) \rangle$, where $\langle \cdot, \cdot \rangle$ is the inner product of the vectors. $\phi(\cdot)$ represents the feature mapping function that maps the original sample to the RKHS. $\mathbf{x}^s, \mathbf{x}^t$ are the instances in source domain D_s and target domain D_t , respectively.

However, considering that the alignment of relevant subdomains within the same category is critical, it can effectively avoid the confusion of irrelevant category distributions. LMMD is a variant of MMD that focuses on the aligning the subdomain distribution of each category from a local perspective. This can be defined as follows.

$$D_{\mathcal{H}}(P_s, P_t) \triangleq \mathbf{E}_{\mathbf{k}} \left\| \mathbf{E}_{\mathbf{x} \sim P_s^{(k)}} [\phi(\mathbf{x}^s)] - \mathbf{E}_{\mathbf{x} \sim P_t^{(k)}} [\phi(\mathbf{x}^t)] \right\|_{\mathcal{H}}^2 \quad (2)$$

where $P_s^{(k)}$ and $P_t^{(k)}$ represent the k th category distribution of D_s and D_t , respectively. Taking Equation (2) as the domain adaptation loss of the deep learning network, through the iterative optimization of the loss function, the distributions of related subdomains in the same class are gradually close to each other. According to the definition of the mean, assuming that the probability corresponding to the category to which each sample belongs is $w^{(k)}$, the unbiased estimation of Eq. (2) can be defined as

$$\hat{D}_{\mathcal{H}}(P_s, P_t) = \frac{1}{K} \sum_{k=1}^K \left\| \sum_{\mathbf{x}_i^s \in D_s} w_i^{s(k)} \phi(\mathbf{x}_i^s) - \sum_{\mathbf{x}_j^t \in D_t} w_j^{t(k)} \phi(\mathbf{x}_j^t) \right\|_{\mathcal{H}}^2 \quad (3)$$

where $w_i^{s(k)}$ and $w_j^{t(k)}$ represent the probability weights of $\mathbf{x}_i^s$ and $\mathbf{x}_j^t$ belonging to the class k , respectively. In addition, $\sum_{i=1}^{N_s} w_i^{s(k)}$ and $\sum_{j=1}^{N_t} w_j^{t(k)}$ are both equal to 1, $\sum_{\mathbf{x}_i \in D} w_i^{(k)} \phi(\mathbf{x}_i)$ denotes the

weighted sum of category k . For each sample $\mathbf{x}_i$, the corresponding $w_i^{(k)}$ calculation equation is as follows:

$$w_i^{(k)} = \frac{y_i^k}{\sum_{(\mathbf{x}_i, \mathbf{y}_i) \in D} y_i^k} \quad (4)$$

where y_i^k is the probability that the sample $\mathbf{x}_i$ belongs to category k . If the sample $\mathbf{x}_i$ is from the source domain D_s , y_i^k can take the k th term of the corresponding one-hot vector label. However, if sample $\mathbf{x}_i$ is from the target domain D_t , it does not have a corresponding one-hot vector label. Considering that classifier $C(\cdot)$ trained by the source domain sample can generate the corresponding category probability distribution for the target sample, this is a suitable method for using the k th term of the probability distribution vector as y_i^k of the target domain sample $\mathbf{x}_i$. One advantage of using probability distribution instead of prediction one-hot label is that probability distributions can avoid the adverse effects of incorrectly predicted labels on the model capability to a certain extent.

In the domain adaptation layer of the depth model, the domain adaptation loss between the target domain D_t and source domain D_s that are relevant, but with different distributions, is defined as follows:

$$\begin{aligned} \hat{D}_{\mathcal{H}}(P_s, P_t) = & \frac{1}{K} \sum_{k=1}^K \left\| \sum_{m=1}^{N_s} \sum_{n=1}^{N_s} w_m^{s(k)} w_n^{s(k)} \mathbf{k}(F(\mathbf{x}_m^s), F(\mathbf{x}_n^s)) \right. \\ & + \sum_{m=1}^{N_t} \sum_{n=1}^{N_t} w_m^{t(k)} w_n^{t(k)} \mathbf{k}(F(\mathbf{x}_m^t), F(\mathbf{x}_n^t)) \\ & \left. - 2 \sum_{m=1}^{N_s} \sum_{n=1}^{N_t} w_m^{s(k)} w_n^{t(k)} \mathbf{k}(F(\mathbf{x}_m^s), F(\mathbf{x}_n^t)) \right\|_{\mathcal{H}}^2 \quad (5) \end{aligned}$$

where $F(\cdot)$ represents the feature extraction function. By minimizing the domain loss in the back-propagation of the deep model, the related subdomain distributions in the same class are gradually become closer to each other.

3. Proposed method

3.1. Network framework of MSSA

In this section, the proposed multi-source information transfer learning model with relevant subdomain distribution alignment is described. As shown in Fig. 2, the basic skeleton of the network model is a one-dimensional CNN, which mainly includes four parts: a shared feature extractor, source-specific feature extractor, source-specific classifier, and task classifier.

Shared feature extractor In order to cut down the network structure and save computing resources, a shared feature extractor $F(x)$ is proposed to project all original signals to a common

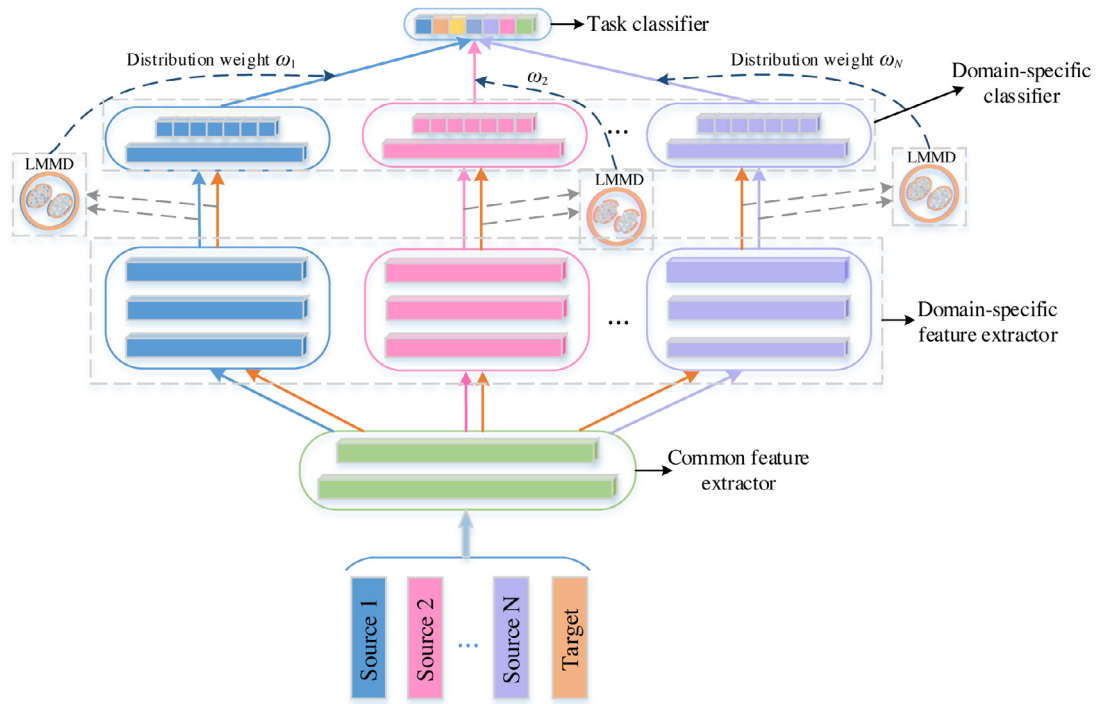


Fig. 2. The network structure of MSSA.

feature space to mine the domain-invariant features of different domains. According to Yosinski, the bottom layer of a neural network is universal feature, and the effect of transfer is better [29]. Therefore, the design of the shared feature extractor is reasonable.

Source-specific feature extractor In order to avoid the difficulty of aligning multiple source and target domains at the same time, a source-specific feature extractor is proposed to project each pair of source domain and target domain to a specific feature space. The relevant subdomain distributions in the same class are aligned in this specific feature space. In particular, for the source-specific extractor $G_j(x)$ that is not shared, the LMMD is applied to reduce the correlation local distribution difference. The domain adaptation loss function can be formulated as

$$l_{LMMD} = \sum_{j=1}^N \hat{D}_{\mathcal{H}}(G_j(F(x^{sj})), G_j(F(x^t))) \quad (6)$$

Source-specific classifier The source-specific classifier $C_j(x)$ is essentially a softmax logistic regression module. It receives the output feature $G_j(F(x))$ from the source-specific feature extractor $G_j(x)$, and outputs the probability distribution of each sample belonging to different categories. Correspondingly, for each source-specific classifier, the cross-entropy loss function is selected to optimize the model parameters. The sum of the classification losses for multiple sources can be calculated as:

$$l_{cls} = \sum_{j=1}^N E_{x \sim X_{sj}} J(C_j(G_j(F(x^{sj}))), y^{sj}) \quad (7)$$

Task classifier Task classifier $C_t(x)$ aims to integrate the results of all specific source classifiers according to the source-specific weight scores. It can synthesize the outputs of all source-specific classifiers to obtain a more reliable diagnosis result. In this study, according to the distribution distance of each pair of source and target domains, the source-specific distribution weight ω_j can be

calculated by Formula (8):

$$\omega_j = \frac{\exp(-\hat{D}_{\mathcal{H}}(G_j(F(x^{sj})), G_j(F(x^t))))}{\sum_{j=1}^N \exp(-\hat{D}_{\mathcal{H}}(G_j(F(x^{sj})), G_j(F(x^t))))} \quad (8)$$

The final diagnosis result can be calculated using Eq. (9), which indicates that the target samples should be classified by multiple source classifiers, and the closer the target domain is to the source domain, the more reliable the classifier prediction result.

$$C_t(x^t) = \sum_{j=1}^N \omega_j \cdot C_j(G_j(F(x^t))) \quad (9)$$

3.2. Optimization objective

A multi-source subdomain adaptation network model for fault diagnosis is proposed in this study, which maps each pair of source and target domains to specific feature spaces to narrow the distribution shift of relevant subdomains, and integrate domain adaptation and feature learning into a deep learning model. First, a shared subnet was applied to extract the shallow common features of all domains in a common feature space. Thereafter, N source-specific subnets that do not share parameters are applied to align the subdomain distribution of each source and target domain. A classifier was learned for each source-specific subnet. Finally, a non-parametric task classifier was constructed to output the final diagnostic results by weighting each source-specific classifier. In summary, the loss of the model includes two parts: LMMD domain adaptation loss and classification loss. By minimizing the LMMD domain adaptation loss, the relevant local distribution differences of each pair of source and target domains can be narrowed. Similarly, the source domain samples can be predicted more accurately by minimizing the classification loss. The total loss is formulated as follows:

$$l_{total} = l_{cls} + \lambda l_{LMMD} \quad (10)$$

where λ trade-off parameter between the two objectives.

Algorithm 1

Input: Labeled source domains $\{D_{sj}\}_{j=1}^N$, unlabeled target domain D_t , initialized F , $G = \{G_j\}_{j=1}^N$ and $C = \{C_j\}_{j=1}^N$

Output: $\hat{Y}_t$, Predicted target sample labels.

while not convergent **do**

1. Randomly sample $\{x_i^{sj}, y_i^{sj}\}_{i=1}^m$ and $\{x_i^t\}_{i=1}^m$ from D_{sj} and D_t .

2. Feed source and target samples to common feature extractor to get the common latent representations $F(x_i^{sj})$ and $F(x_i^t)$.

3. Feed $F(x_i^{sj})$ and $F(x_i^t)$ to domain-specific feature extractor to get domain-specific representations $G_j(F(x_i^{sj}))$ and $G_j(F(x_i^t))$.

4. Feed $G_j(F(x_i^{sj}))$ to domain-specific classifier to get $C_j(G_j(F(x_i^{sj})))$.

5. Use $G_j(F(x_i^{sj}))$ and $G_j(F(x_i^t))$ to calculate the domain adaptation loss as Equation (6).

6. Use $C_j(G_j(F(x_i^{sj})))$ and y_i^{sj} to compute the classification loss as Equation (7).

7. Update the common feature extractor F , multiple domain-specific feature extractor $\{G_j\}_{j=1}^N$ and multiple domain-specific classifier $\{C_j\}_{j=1}^N$ by minimizing the total loss in Equation (10).

end while

Based on the proposed network model, the parameters of the shared and the source-specific feature extractors can be optimized by simultaneously minimizing l_{cls} and l_{LMMD} . The source-specific classifier parameters can be optimized by minimizing l_{cls} . The formulation of the optimization problem is:

$$\begin{aligned}\hat{\theta}_F &= \arg\{\min_{\theta_F} l_{cls}(\theta_F, \hat{\theta}_G, \hat{\theta}_C), \min_{\theta_F} l_{LMMD}(\theta_F, \hat{\theta}_G, \hat{\theta}_C)\} \\ \hat{\theta}_G &= \arg\{\min_{\theta_G} l_{cls}(\hat{\theta}_F, \theta_G, \hat{\theta}_C), \min_{\theta_G} l_{LMMD}(\hat{\theta}_F, \theta_G, \hat{\theta}_C)\} \\ \hat{\theta}_C &= \arg\{\min_{\theta_C} l_{cls}(\hat{\theta}_F, \hat{\theta}_G, \theta_C)\}\end{aligned}\quad (11)$$

where θ_F , θ_G , and θ_C represent the learnable parameters of the shared feature extractor, source-specific feature extractor, and source-specific classifier, respectively. $\hat{\theta}_F$, $\hat{\theta}_G$, and $\hat{\theta}_C$ are the optimal values of θ_F , θ_G and θ_C , respectively.

The weight parameters of network model can be updated in each iteration training:

$$\begin{aligned}\theta_F &\leftarrow \theta_F - \eta \left(\frac{\partial l_{cls}}{\partial \theta_F} + \lambda \frac{\partial l_{LMMD}}{\partial \theta_F} \right) \\ \theta_G &\leftarrow \theta_G - \eta \left(\frac{\partial l_{cls}}{\partial \theta_G} + \lambda \frac{\partial l_{LMMD}}{\partial \theta_G} \right) \\ \theta_C &\leftarrow \theta_C - \eta \frac{\partial l_{cls}}{\partial \theta_C}\end{aligned}\quad (12)$$

where η denotes the learning rate. The parameter optimizations in Eq. (12) can be accomplished by the stochastic gradient descent (SGD) algorithm. The entire process of the proposed framework is summarized in Algorithm 1.

4. Experimental study

To verify the superiority of the proposed MSSA in this section, two datasets are collected from two different rotating bearing experimental platforms for extensive experiments, and several state-of-the-art methods were introduced for performance comparison with the proposed method.

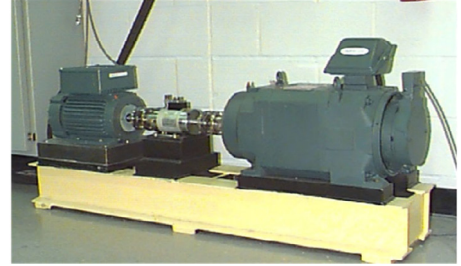


Fig. 3. Test rig of CWRU dataset.

4.1. Dataset descriptions

(1) CWRU dataset: The CWRU dataset is a public bearing fault dataset provided by Case Western Reserve University [30]. The test rig is shown in Fig. 3. The instrument consists of a 2 HP motor, dynamometer, and torque transducer. The two acceleration sensors were placed on the fan end and drive end of the motor for vibration signal collection. Three types of faults: outer race fault (ORF), ball fault (BF), and inner race fault (IRF) are introduced through electro-discharge machining. Each fault type contains three fault diameters of 0.021 inch, 0.014 inch, and 0.007 inch. With the addition of normal bearings, there were a total of ten health states. Table 1 lists the detailed status information. In the experiment, we selected the drive end vibration signals under four different speed conditions (1730 rpm, 1750 rpm, 1772 rpm and 1797 rpm) as the experimental data, and all the data were obtained at a sampling frequency of 12 kHz.

(2) QPZZ-II dataset: The QPZZ-II dataset was collected from a rotating machinery vibration analysis and fault detection test rig, as shown in Fig. 4(a). An AC variable-frequency motor with a power of 0.75 kW was utilized to provide the driving force, and a manual loading device was used to change the load. The bearing

Table 1
Description of CWRU bearing health conditions.

Condition	Normal	IRF	IRF	IRF	BF	BF	BF	ORF	ORF	ORF
Severity (inch)	/	0.007	0.014	0.021	0.07	0.014	0.021	0.007	0.014	0.024
Label	0	1	2	3	4	5	6	7	8	9

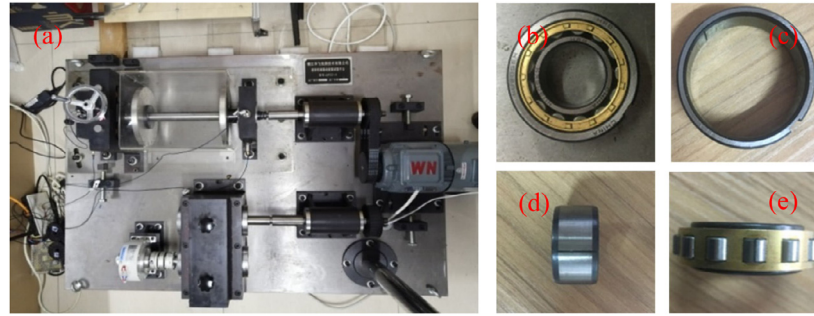


Fig. 4. (a) Test rig for vibration analysis and fault diagnosis. (b) Normal. (c) Outer race fault. (d) Inner race fault. (e) Roller fault.

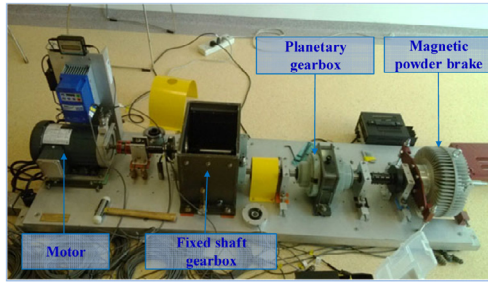


Fig. 5. Test rig of Gearbox dataset.

data of four different states (health, BF, ORF, IRF) were collected under loads of 0, 1, and 2. The single point of fault of the bearing is introduced through artificial processing, as shown in Fig. 4(b)–(e). The motor speed was 1450 rpm and the sampling frequency was 10240 Hz.

(3) Gearbox dataset: The gearbox dataset was collected on an experimental platform as shown in Fig. 5. The main components of this experimental platform include an AC variable-frequency motor, planetary gearbox, fixed-shaft gearbox, and magnetic powder brake for applying different loads. During data collection, the planetary gearbox was always in a normal state, and the fixed-shaft gearbox sequentially replaced gears with different fault modes to simulate different mechanical states. The magnetic powder brake control current is set to 0.2 A, 0.1 A and 0.05 A to apply different levels of load. The motor speed was controlled at 880 rpm and the sampling frequency was 5120 Hz. The fault forms include a single-point fault of the large gear, small gear, and compound fault of both, and the detailed health conditions of the fixed-shaft gearbox are listed in Table 2.

4.2. Compared methods

To highlight the advantages of the developed MSSA, several popular approaches were used in this experiment for comparison with MSSA. For a fair comparison, these methods adopt a similar network structure and hyperparameters as MSSA. For the single-source domain adaptation (SUDA) method, only the best diagnostic result was provided.

(1) Supervised CNN: Supervised CNN is a traditional fault diagnosis method without domain adaptation. It uses a single source data for training, and directly tests the target domain.

(2) DAN with a single source [10]: DAN is a deep domain adaptation network that uses MMD to measure the feature distances at multiple high levels between the target and source domains, and achieves distribution alignment by minimizing the domain difference loss.

(3) JAN with a single source [11]: JAN is deep transfer learning with a joint adaptation network, which applies an improved MMD to match cross-domain joint distribution differences.

(4) DANN with a single source [31]: DANN is a domain adversarial network that generates domain-invariant representations through a game between domain feature generators and domain discriminators.

(5) MCD with a single source [32]: MCD is a UDA method with maximum classifier discrepancy that attempts to align the distributions of the source and target by maximizing the discrepancy between the outputs of the two classifiers.

(6) DAN-LMMD with a single source: DAN-LMMD is a deep domain adaptation network for domain measurement that replaces the MMD difference measurement method in DAN with LMMD.

(7) DAN with source combination: Constructing a single data source by integrating different distributed source domains and applying DAN for domain adaptation.

(8) DANN with source combination: Constructing a single data source by integrating different distributed source domains and applying DANN for domain adaptation.

(9) DAN-LMMD with source combination: Constructing a single data source by integrating different distributed source domains and applying LMMD for domain adaptation.

(10) ADACL [33]: ADACL is a multi-source adversarial domain adaptation method with classifier alignment for cross-domain intelligent fault diagnosis.

(11) MADN [34]: MADN is a stacked autoencoder-based multi-source adaptation deep transfer learning network for industrial process fault diagnosis.

It should be noted that for both the DAN-LMMD and DAN methods, the distance measurement was performed on the last three feature layers. For the proposed MSSA, distance measurement in multiple feature layers did not significantly improve the model diagnosis performance. Therefore, to save computing resources in MSSA, only the last layer of the source-specific feature extractor is applied with LMMD to measure the distribution distance.

Table 2

Description of the health conditions of the fixed-shaft gearbox.

Label	Condition
0	Normal
1	Large gear pitting
2	Large gear tooth broken
3	Large gear tooth broken and small gear wear
4	Large gear pitting and small gear wear
5	Small gear wear

Table 3

Description of source domain data.

	Source domain name	Working condition
CWRU dataset	A	Rotating speed : 1797 rpm
	B	Rotating speed : 1772 rpm
	C	Rotating speed : 1750 rpm
	D	Rotating speed : 1730 rpm
QPZZ-II dataset	E	load 0
	F	load 1
	G	load 2
Gearbox dataset	H	Control current: 0.2 A
	I	Control current: 0.1 A
	J	Control current: 0.05 A

4.3. Experimental setup

For the data collected on the three experimental platforms, the vibration data under different working conditions were respectively regarded as a source domain, as shown in Table 3. For the vibration signal in each health state, a sliding window of size 1024 was used to intercept 300 samples. Therefore, for the CWRU dataset, 3000 samples were collected for each source, and for the QPZZ-II dataset, 1200 samples were collected for each source. Similarly, each source collected 1800 samples for the gearbox dataset. In the trial, one source was used as the target domain in turn, and the rest was used as multiple source domains. Seventy percent of the samples were randomly selected from each source as the training set, and the remaining 30% were used as the testing set. Each trial was repeated five times, and the average was taken as the final diagnosis result to avoid randomness. All experiments were implemented using PyCharm software through PyTorch on a computer with 2.9 GHz i7-10700F processor and 8 GB RAM.

The specific network architecture parameters of the proposed MSSA are presented in Table 4. To reasonably process one-dimensional vibration signals with a large data length, a larger convolution kernel was applied to the first convolution layer of the shared feature extractor, and the remaining convolution layers used a small convolution kernel with a size of 1×3 . In addition, Rectified linear unit (ReLU) was applied as the activation function of the entire network. The trade-off parameter λ is gradually changed from 0 to 1 using the formula $2/\exp(-10 \times p) - 1$, to reduce the influence of parameter sensitivity, where p denotes the training progress linearly changing from 0 to 1. The network learning rate was set to 0.01, and 10 epochs were conducted in one training.

4.4. Results discussion and analysis

4.4.1. Comparison and analysis with SUDA methods

In this experiment, the mini-batch size was set to 64 for all comparison methods. The diagnosis results of the proposed method and several SUDA methods on the CWRU dataset are shown in Fig. 6 and Table 4. As shown in the results, the proposed MSSA has a significant improvement in diagnostic accuracy and stability compared with the SUDA methods for different

Table 4

The network architecture parameters of the proposed MSSA. Conv, MP, BN, FC, DR stand for Convolutional layer, Max Pooling, batch normalization, fully connected layer, and Dropout layers respectively. Conv($m \times (1 \times n)$) means that this layer has m convolution kernels with a size of $1 \times n$.

Network component	Network structure parameter
Input	Input layer (1024)
Shared feature extractor	Layer1: Conv($16 \times (1 \times 32)$) BN(16) MP(1×2) Layer2: Conv($32 \times (1 \times 3)$) BN(32) MP(1×2)
Source-specific feature extractor	Layer1: Conv($64 \times (1 \times 3)$) BN(64) MP(1×2) Layer2: Conv($128 \times (1 \times 3)$) BN(128) MP(1×2) Layer3: Conv($128 \times (1 \times 3)$) BN(128) MP(1×2)
Source-specific classifier	Layer1: FC(128) BN(128) DR(0.3) Layer2: FC(k)
Task classifier	Output layer (k)

diagnostic tasks. More specifically, the traditional CNN without domain adaptation achieved the lowest diagnosis accuracy, with an average accuracy of only 87.66% for four different diagnostic tasks. The introduction of domain adaptation effectively improves the diagnostic performance of the model. As shown in Table 5, DAN, JAN, DANN, MCN, and DAN-LMMD increased the average diagnosis accuracies to 93.99%, 94.82%, 94.62%, 94.90%, and 96.25%, respectively. The average diagnostic accuracy of DAN-LMMD was improved by 2.26%, 1.43%, 1.63%, and 1.35% compared to DAN, JAN, DANN, and MCN, respectively. The improvement from global domain adaptation to subdomain adaptation is the key to the increase in diagnostic accuracy, and LMMD can accurately align the distribution of relevant subdomains and capture more fine-grained information for each category. In addition, MSSA improves the average diagnosis accuracy by 3.62% based on DAN-LMMD, which shows that more abundant related diagnosis knowledge can be learned from multiple source domains, which is beneficial for improving the generalization ability of the model.

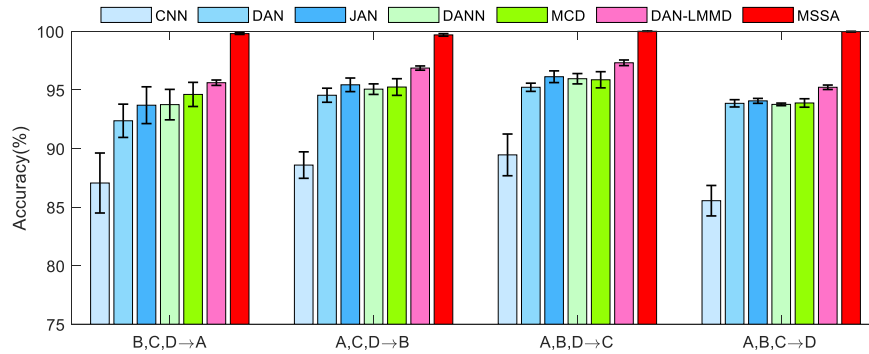
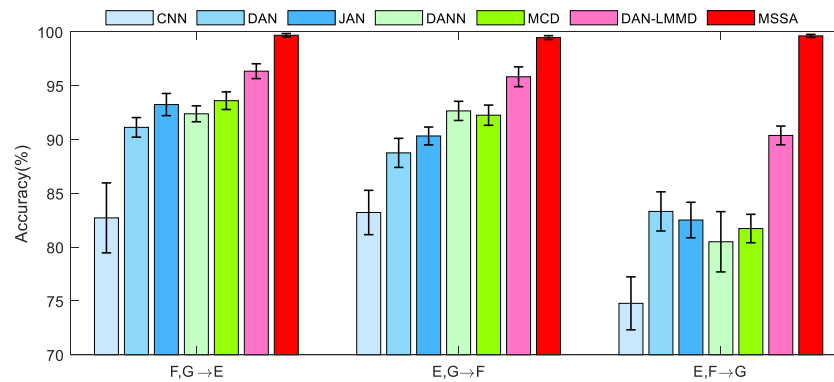
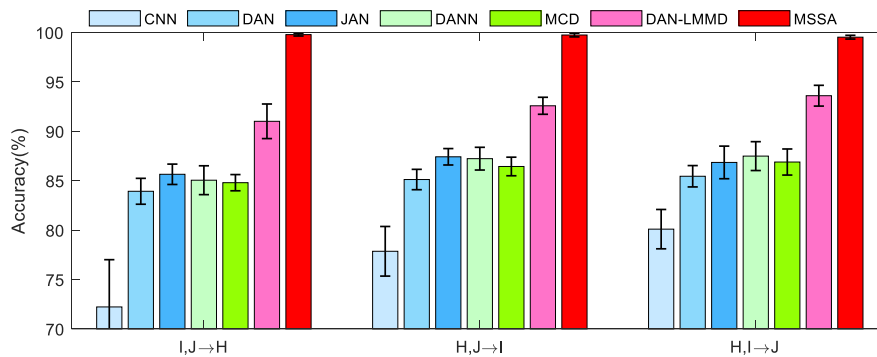
The diagnosis results of the proposed method and several SUDA methods on the QPZZ-II dataset are displayed in Fig. 7 and Table 6. Similar diagnostic results were achieved as those on the CWRU dataset. It is noteworthy that the diagnostic performance of several single source domain adaptation methods for diagnostic task E, F $\rightarrow$ G is poor, while the accuracy of DAN-LMMD can still reach 90.38%, and the proposed MSSA still maintains a high diagnostic accuracy of 99.63%, which again verifies the effectiveness of fine-grained information matching and multi-source knowledge transfer strategies. The diagnosis results of the different methods on the Gearbox dataset are presented in Fig. 8 and Table 7. It can be seen that the diagnostic performance of the proposed MSSA is significantly greater than several other SUDA methods. This may be because the existence of compound fault forms makes the features of different fault forms more similar, and the SUDA diagnosis model makes it difficult to distinguish clear decision-making boundaries in high-level feature space. The proposed MSSA utilizes LMMD in each specific source feature extractor to realize fine-grained information migration, which effectively improves diagnosis accuracy.

Furthermore, non-parametric statistical tests were applied for significance analysis between the different diagnostic methods. According to [35], the Friedman test can compare multiple classifiers on multiple datasets; therefore, it is adopted in this study. The null hypothesis is that all diagnostic methods have no significant differences and that their ranks should be equal. Table 8 presents the average ranks of the results of the different methods for each diagnostic task. According to the formula in [35], the test statistic was calculated as $F_F = 69.72$. With seven approaches and 10 diagnostic tasks, F_F obeys the F distribution with $7 - 1 = 6$ and $(7 - 1) \times (10 - 1) = 54$ degrees of freedom. Under the condition of significance level $\alpha = 0.05$, the critical value of

Table 5

Average diagnostic accuracy and standard deviation of different methods on the CWRU dataset.

Method	B,C,D→A	A,C,D→B	A,B,D→C	A,B,C→D	Average
CNN	87.05 ± 2.56	88.58 ± 1.13	89.45 ± 1.78	85.54 ± 1.30	87.66
DAN	92.36 ± 1.42	94.54 ± 0.60	95.22 ± 0.35	93.85 ± 0.31	93.99
JAN	93.69 ± 1.57	95.43 ± 0.58	96.12 ± 0.50	94.06 ± 0.21	94.82
DANN	93.74 ± 1.30	95.06 ± 0.45	95.95 ± 0.44	93.76 ± 0.11	94.62
MCD	94.61 ± 1.03	95.24 ± 0.71	95.86 ± 0.69	93.88 ± 0.36	94.90
DAN-LMMD	95.61 ± 0.23	96.86 ± 0.18	97.31 ± 0.24	95.22 ± 0.19	96.25
MSSA	99.80±0.07	99.69±0.11	100±0	99.97±0.03	99.87

**Fig. 6.** Diagnostic accuracy on the CWRU dataset with different methods.**Fig. 7.** Diagnostic accuracy on the QPZZ-II dataset with different methods.**Fig. 8.** Diagnostic accuracy on the Gearbox dataset with different methods.

$F(6, 54)$ is 2.27, which is much smaller than the test statistic; thus, the null hypothesis is rejected. A post-hoc test was further applied to compare the other approaches with the proposed MSSA. Calculated using the formula in [35], the critical value for the significant difference in diagnostic performance between the two algorithms is 2.55. The results of the data analysis in Table 8 are shown in Fig. 9. The methods on the right side of the axis are better, and all the methods with ranks outside

the marked interval are significantly different from the proposed MSSA. Therefore, compared with the traditional SUDA method, the proposed MSSA exhibited a significant improvement in diagnostic performance.

4.4.2. Comparison and analysis with MUDA methods

In addition to the comparison with the SUDA method, several multi-source domain adaptation fault diagnosis methods are

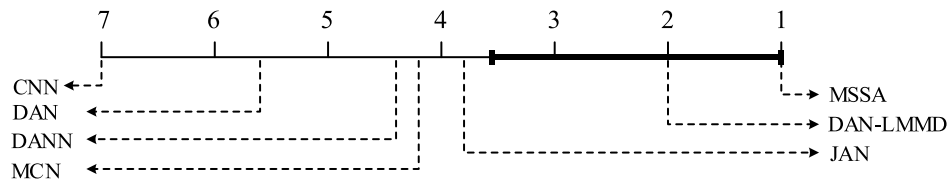


Fig. 9. Comparison of proposed MSSA against other methods.

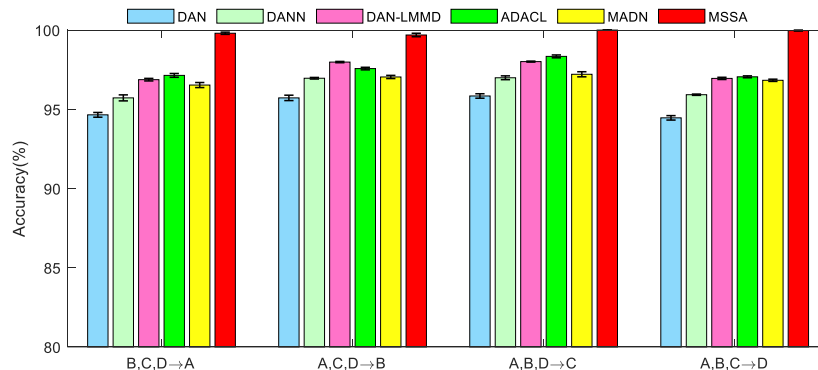


Fig. 10. Diagnostic accuracy on the CWRU dataset with different methods.

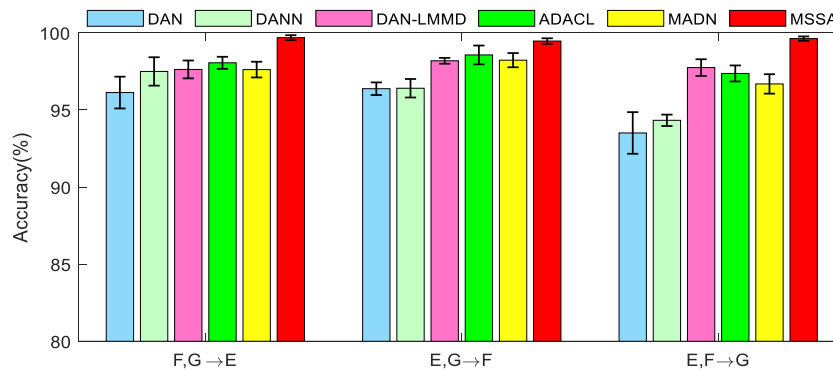


Fig. 11. Diagnostic accuracy on the QPZZ-II dataset with different methods.

Table 6

Average diagnostic accuracy and standard deviation of different methods on the QPZZ-II dataset.

Method	F,G→E	E,G→F	E,F→G	Average
CNN	82.72 ± 3.25	83.22 ± 2.06	74.78 ± 2.46	80.24
DAN	91.13 ± 0.91	88.76 ± 1.35	83.32 ± 1.82	87.74
JAN	93.25 ± 1.03	90.33 ± 0.83	82.52 ± 1.65	88.70
DANN	92.39 ± 0.74	92.66 ± 0.89	80.50 ± 2.80	88.52
MCD	93.61 ± 0.82	92.26 ± 0.94	81.73 ± 1.32	89.20
DAN-LMMD	96.35 ± 0.69	95.83 ± 0.92	90.38 ± 0.87	94.19
MSSA	99.69 ± 0.16	99.47 ± 0.18	99.63 ± 0.14	99.60

Table 7

Average diagnostic accuracy and standard deviation of different methods on the Gearbox dataset.

Method	I,J→H	H,J→I	H,I→J	Average
CNN	72.22 ± 4.78	77.85 ± 2.51	80.09 ± 1.99	76.72
DAN	83.92 ± 1.31	85.11 ± 1.03	85.44 ± 1.08	84.82
JAN	85.64 ± 1.12	87.41 ± 0.92	86.84 ± 1.56	86.63
DANN	85.04 ± 1.46	87.22 ± 1.15	87.48 ± 1.46	86.58
MCD	84.79 ± 0.86	86.43 ± 1.02	86.88 ± 1.39	86.03
DAN-LMMD	91.00 ± 1.75	92.57 ± 0.86	93.59 ± 1.05	92.39
MSSA	99.77 ± 0.14	99.73 ± 0.19	99.51 ± 0.19	99.67

Table 8

The average ranks of the accuracy for each diagnostic task by different methods.

	CNN	DAN	JAN	DANN	MCD	DAN-LMMD	MSSA
Average rank	7.00	5.60	3.80	4.40	4.20	2.00	1.00

introduced in this section for comparison with the proposed MSSA. Similarly, the mini-batch size was set to 64 for all comparison methods. The diagnosis results of the proposed method and several MUDA methods on the CWRU dataset are shown in Fig. 10 and Table 9. Compared with the SUDA method, the average diagnostic precision of DAN, DANN, and DAN-LMMD of the source combination increased by 1.18%, 1.78%, and 1.20%, respectively, which is limited in terms of improving diagnostic performance. However, the proposed MSSA obtains a better diagnostic performance of 99.87%, which demonstrates that it is advantageous to learn multiple domain-invariant representations from each pair of source and target domains, whereas it is difficult to extract common domain-invariant representations from all domains. The diagnostic results of ADACL and MADN were 97.52% and 96.90%, respectively, similar to those of DAN-LMMD. This is because although ADACL and MADN respectively use the domain adaptation method of multiple classifier alignment and multiple domain recognizer adversarial training, in essence, they all use

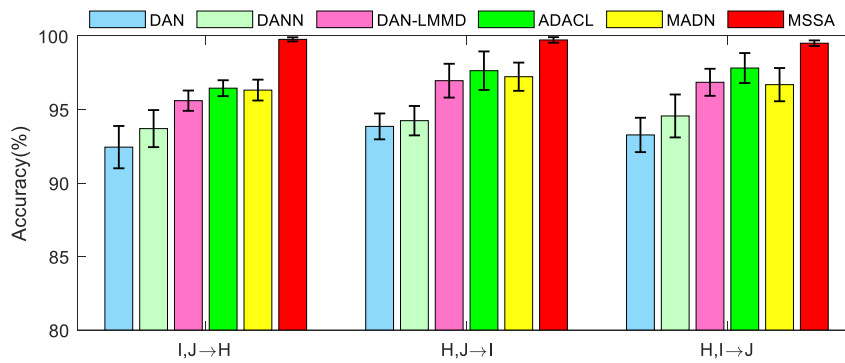


Fig. 12. Diagnostic accuracy on the Gearbox dataset with different methods.

Table 9

Average diagnostic accuracy and standard deviation of different methods on the CWRU dataset.

Method	B,C,D→A	A,C,D→B	A,B,D→C	A,B,C→D	Average
DAN	94.65 ± 0.15	95.72 ± 0.17	95.84 ± 0.14	94.46 ± 0.14	95.17
DANN	95.72 ± 0.19	96.96 ± 0.05	96.99 ± 0.12	95.92 ± 0.04	96.40
DAN-LMMD	96.87 ± 0.08	97.98 ± 0.04	98.01 ± 0.03	96.95 ± 0.07	97.45
ADACL	97.14 ± 0.12	97.57 ± 0.08	98.34 ± 0.09	97.05 ± 0.06	97.52
MADN	96.53 ± 0.16	97.04 ± 0.10	97.21 ± 0.16	96.83 ± 0.07	96.90
MSSA	99.80±0.07	99.69±0.11	100±0	99.97±0.03	99.87

Table 10

Average diagnostic accuracy and standard deviation of different methods on the QPZZ-II dataset.

Method	F,G→E	E,G→F	E,F→G	Average
DAN	96.13 ± 1.03	96.38 ± 0.41	93.51 ± 1.35	95.34
DANN	97.50 ± 0.92	96.41 ± 0.60	94.33 ± 0.37	96.08
DAN-LMMD	97.63 ± 0.58	98.19 ± 0.19	97.75 ± 0.54	97.86
ADACL	98.06 ± 0.39	98.57 ± 0.61	97.37 ± 0.52	98.00
MADN	97.62 ± 0.51	98.23 ± 0.46	96.69 ± 0.63	97.51
MSSA	99.69±0.16	99.47±0.18	99.63±0.14	99.60

a shared feature extractor to map all source and target domains to the same feature space for distribution matching, which is a difficult task.

The diagnosis results of the proposed method and several MUDA methods on the QPZZ-II dataset are shown in Fig. 11 and Table 10. Similarly, the proposed MSSA achieved the highest accuracy for different diagnostic tasks. It is worth noting that the multi-source domain adaptation diagnosis methods exhibit a slight improvement in diagnostic performance compared with the SUDA methods. Compared with the SUDA method, the diagnostic precision of DAN, DANN, and DAN-LMMD of the source combination increased by 7.60%, 7.56%, and 3.67%, respectively, indicating that the diagnostic knowledge of a single source domain is very limited in some diagnostic tasks, and knowledge transfer from multiple source domains can effectively improve model diagnostic performance. Fig. 12 and Table 11 show the diagnosis results of the different MUDA models on the gearbox dataset. Although there is a certain degree of improvement, the source domain combination method still cannot align the distribution of multiple source domains and target domains well, and the enhancement of the diagnosis result is limited. DAN-LMMD, ADACL, and MADN achieved an approximate diagnostic accuracy, and the average diagnostic accuracy of the proposed MSSA on the three diagnostic tasks exceeded that of the most competitive ADACL by 2.37%. In addition, based on standard deviation of the different SUDA methods, the proposed MSSA exhibited better stability.

Table 11

Average diagnostic accuracy and standard deviation of different methods on the Gearbox dataset.

Method	I,J→H	H,J→I	H,I→J	Average
DAN	92.44 ± 1.44	93.85 ± 0.88	93.27 ± 1.17	93.19
DANN	93.70 ± 1.26	94.24 ± 1.00	94.56 ± 1.46	94.17
DAN-LMMD	95.60 ± 0.69	96.96 ± 1.15	96.85 ± 0.92	96.47
ADACL	96.45 ± 0.54	97.64 ± 1.31	97.82 ± 1.02	97.30
MADN	96.32 ± 0.71	97.24 ± 0.96	96.67 ± 1.13	96.74
MSSA	99.77±0.14	99.73±0.19	99.51±0.19	99.67

Table 12

The average ranks of the accuracy for each diagnostic task by different methods.

	DAN	DANN	DAN-LMMD	ADACL	MADN	MSSA
Average rank	6.00	5.00	3.10	2.20	3.70	1.00

The average ranks of the results of the different MUDA methods for each diagnostic task are displayed in Table 12. With six approaches and 10 diagnostic tasks, the test statistics can be calculated as $F_F = 173.93$, and the critical value $F(6, 45) = 2.30$ with a significance level $\alpha = 0.05$ by querying the F-test critical value table. The calculated test statistics are much larger than the critical value; therefore, there are significant differences in the diagnostic performance of the different MUDA methods. In the post-hoc test, the critical value of the significant difference between the two methods was calculated as 2.16. The results of the data analysis in Table 12 are presented in Fig. 13. It can be observed that the proposed MSSA is significantly superior to the source combination method and it outperforms the advanced ADACL by a certain amount.

4.4.3. Model performance analysis

To verify the availability of the relevant subdomain alignment of MSSA, t-distributed stochastic neighbor embedding (t-SNE) [36] technology was applied to visualize the feature representation extracted by the source-specific feature extractor. The feature visualizations of the supervised CNN and MSSA on the CWRU dataset are displayed in Fig. 14. As shown in Fig. 14(a), there is

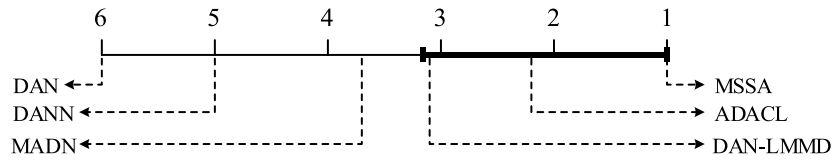


Fig. 13. Comparison of proposed MSSA against other methods.

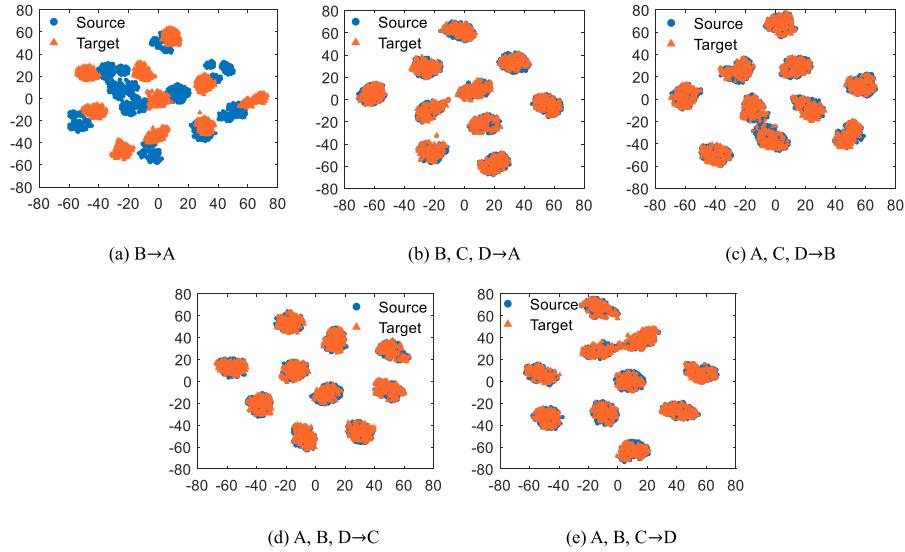


Fig. 14. Feature visualization on the CWRU dataset. (a) Supervised CNN, (b)–(e) MSSA.

a significant discrepancy between the source domain and target domain features in a supervised CNN without domain adaptation, which indicates that the training model based only on the source domain cannot be well generalized to the target domain. Fig. 14(b)–(e) show the feature representation visualization of one of the source and target domains in different cross-domain diagnostic tasks. It can be seen that the features of the related categories between the two domains are well aligned, which demonstrates the excellent characteristics of the LMMD in narrowing the distribution difference of the related subdomains. The results of the feature visualization on the QPZZ-II dataset are illustrated in Fig. 15. A feature distribution pattern similar to that of the CWRU dataset was obtained, demonstrating the outstanding generalization ability of the proposed MSSA on different datasets.

In addition, to further demonstrate the diagnostic precision of MSSA for each category, Figs. 16 and 17 display the diagnostic confusion matrix of MSSA on the CWRU and QPZZ-II datasets, respectively. As shown in Fig. 16, on the CWRU dataset, only one type of fault state in different diagnosis tasks is classified incorrectly, and the rest of the health states are classified correctly. Especially in A, B, D→C task, 100% diagnostic accuracy was obtained for each class. Similarly, as shown in Fig. 17, on the QPZZ-II dataset, most of the fault categories in the different tasks were all classified correctly, and the lowest classification accuracy reached 98.89%. In general, the proposed MSSA achieves superior diagnosis results for each fault category in different diagnosis tasks on the two datasets and verifies the prominent generalization capability of MSSA in cross-domain fault diagnosis.

The distribution distance weights learned from specific sources in different diagnostic tasks are shown in Figs. 18 and 19. Generally, data sources collected from machines with little change in operating conditions have relatively small distribution distances, and correspondingly larger weights can be obtained. It can be

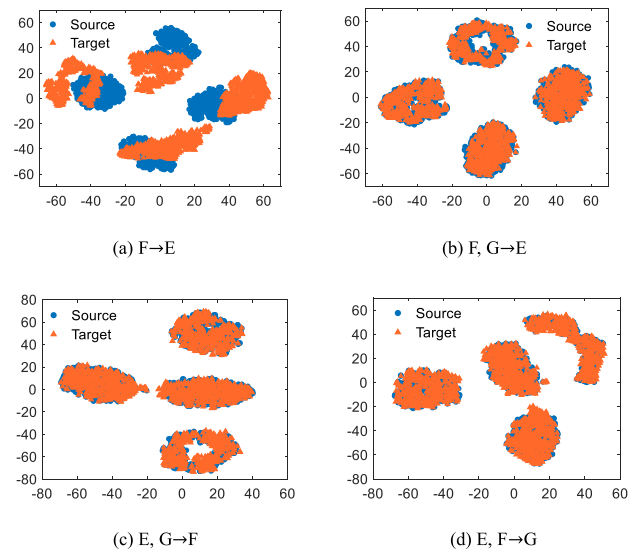


Fig. 15. Feature visualization on the QPZZ-II dataset. (a) Supervised CNN, (b)–(e) MSSA.

observed that as the number of training epochs increases, the learned source-specific weight gradually stabilizes. Specifically, the source domain, whose working condition is closer to the target domain, obtains more weight, and vice versa. On the CWRU dataset, the weight learned between sources A and B greater than the weight learned from other sources, as shown in Fig. 18(a) and (b). Similarly, Fig. 18(c) and (d) demonstrate that a relatively large weight has been learned between sources C and D, which is consistent with the small difference in mechanical rotation speed between them. On the QPZZ-II dataset, the source domain whose

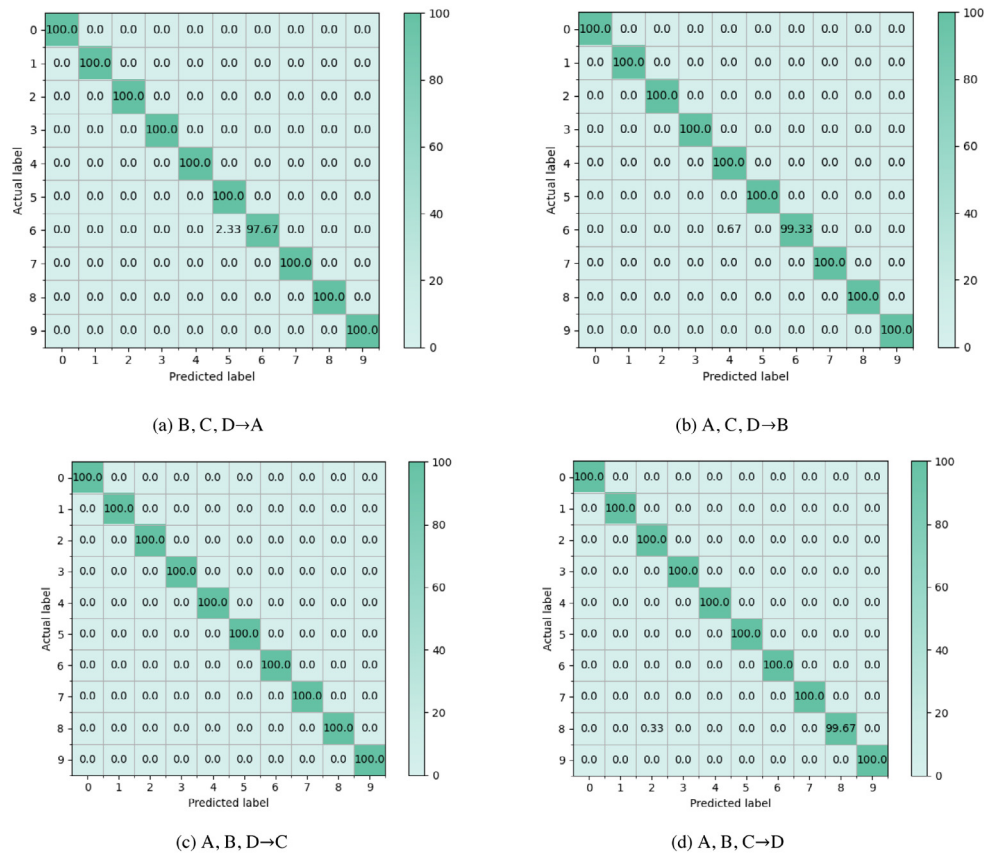


Fig. 16. Confusion matrix of the proposed MSSA on the CWRU dataset (%).

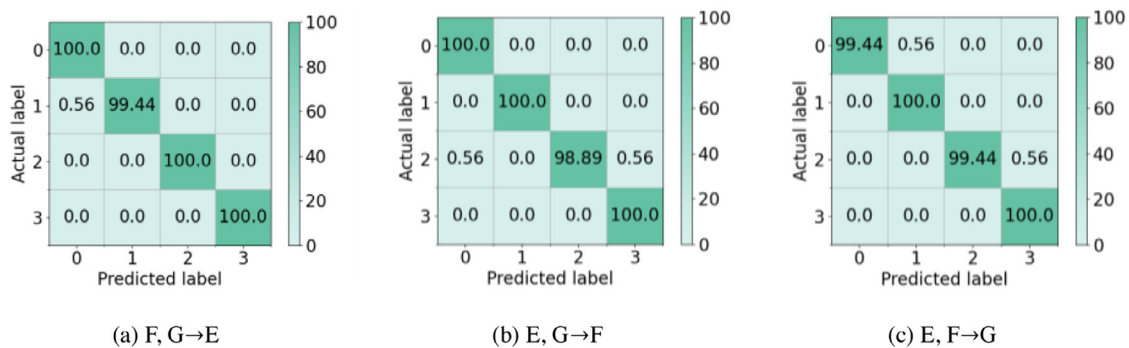


Fig. 17. Confusion matrix of the proposed MSSA on the QPZZ-II dataset (%).

operating load is closer to the target domain is assigned greater weight in the final diagnosis result. It is worth mentioning that in the diagnostic task E, $G \rightarrow F$, the two source domains maintain a balanced load gap with the target domain, therefore, approximate weights are obtained.

4.4.4. Comparison of different training sample sizes in the target domain

In actual fault diagnosis scenarios, collecting various fault data is a resource-intensive task. An ideal model should use as few training samples as possible to obtain a high diagnostic accuracy. Therefore, in this experiment, taking the B, C, D→A task on the CWRU dataset as an example, 70% of the source domain data and different proportions of target domain data were applied to verify the diagnostic capability of MSSA in the training mode with fewer target domain samples. To ensure that the model was fully

trained, the mini-batch was set to 30 in this experiment to train the model more iteratively.

Fig. 20 and Table 13 show the diagnosis results obtained using different proportions of the target domain training samples. It can be seen that MSSA always maintains the highest diagnostic precision compared with the other methods in the diagnostic trial of the proportion of each training sample. In addition, as the proportion of the target domain training samples decreased, the diagnostic accuracy of each model decreased. Interestingly as the sample size decreases, the diagnosis precision of the domain adaptation methods based on distance measurement decreases slightly. The diagnostic accuracy of individual methods increased slightly, even when the sample proportion was reduced. For example, the DAN-LMMD with single source, when the target domain training sample proportion is 50% and 30%, the diagnostic accuracy is 94.35% and 95.12% respectively, which shows that

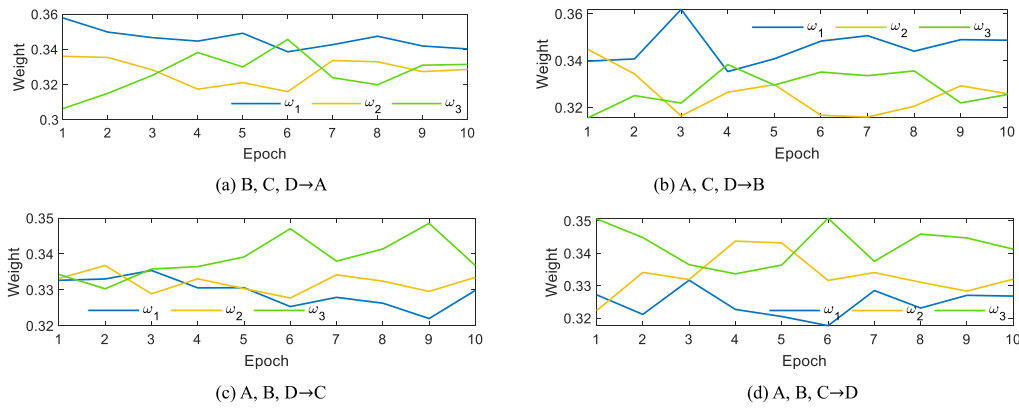


Fig. 18. Source-specific weights learned on the CWRU dataset.

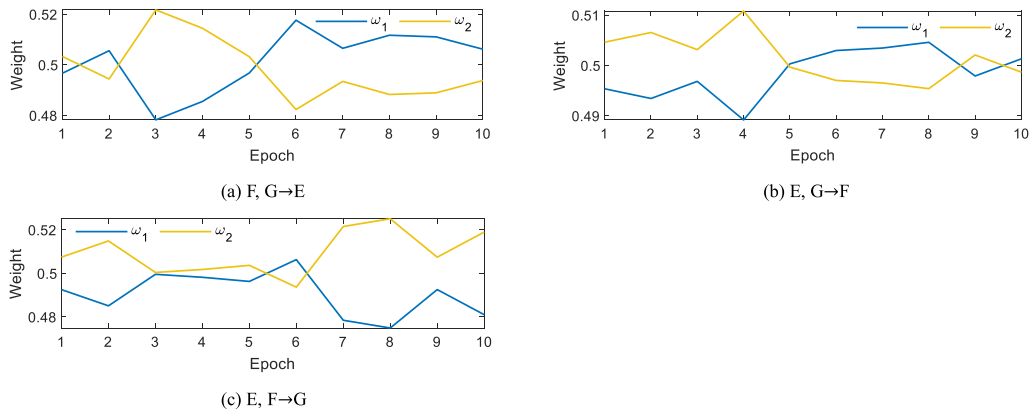


Fig. 19. Source-specific weights learned on the QPZZ-II dataset.

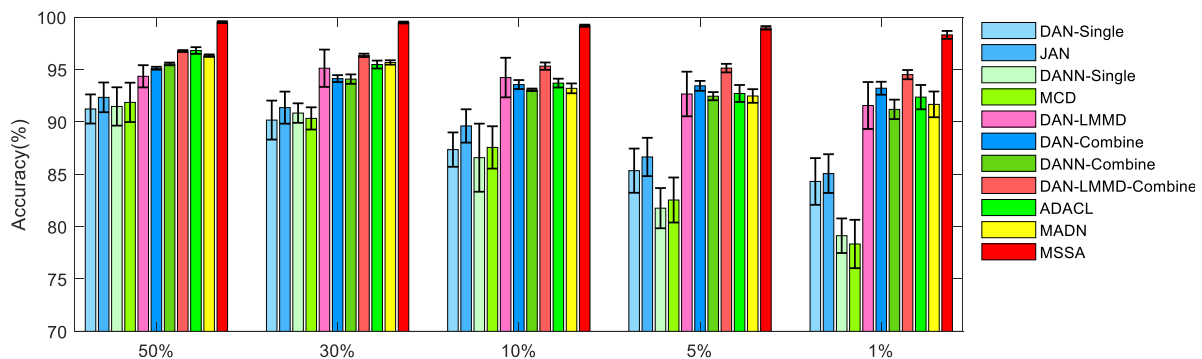


Fig. 20. Diagnostic accuracy on the CWRU dataset with different proportions of target domain samples.

the shift domain alignment can be achieved with fewer training samples. Moreover, the domain adaptation method based on adversarial training and classifier alignment exhibits a significant decrease in diagnostic accuracy as the number of training samples decreases. When the training sample size is small, DANN-Single, MCN, DANN-Combine ADACL, and MADN have relatively low diagnostic accuracies, which indicates that approaches based on the adversarial strategy and classifier alignment are more sensitive to the training sample size, whereas approaches based on distance measurement are more robust to the sample size change. The proposed MSSA based on subdomain alignment can achieve more

fine-grained measurement than global domain alignment; therefore, the proposed method can still achieve 98.30% diagnostic accuracy when the target domain sample proportion is only 1%.

4.4.5. Comparison of imbalanced training sample in the target domain

Another real scenario in the field of fault diagnosis is that the number of unlabeled training samples collected in the target domain is seriously imbalanced in each category. It is often easier to collect the mechanical vibrations of signals in the normal state, and signals in the fault state are relatively scarce. Therefore, in order to test the diagnostic performance of the proposed model

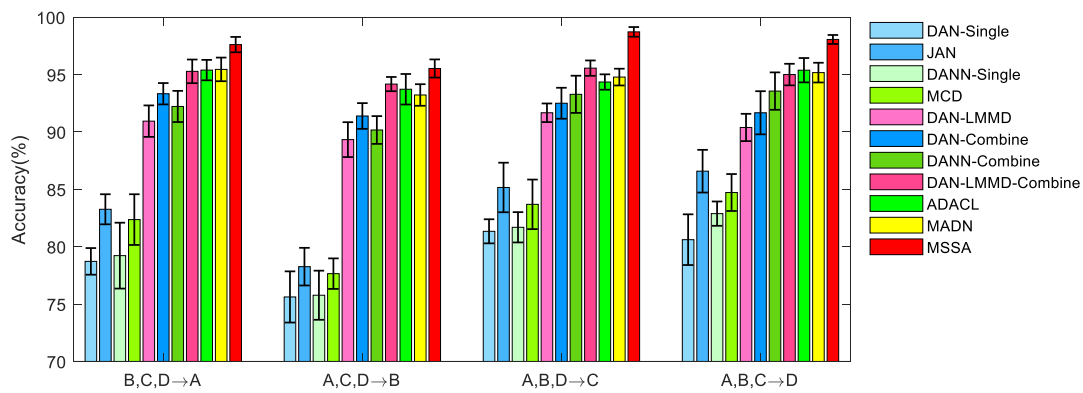


Fig. 21. Diagnostic accuracy on the CWRU dataset with imbalanced training sample in the target domain.

Table 13

Average diagnostic accuracy and standard deviation of different methods on the CWRU dataset with different proportions of target domain samples.

Standards	Method	50%	30%	10%	5%	1%
Single Best	DAN	91.23 ± 1.39	90.17 ± 1.86	87.35 ± 1.64	85.34 ± 2.11	84.31 ± 2.23
	JAN	92.34 ± 1.42	91.36 ± 1.53	89.61 ± 1.59	86.65 ± 1.83	85.06 ± 1.83
	DANN	91.47 ± 1.83	90.84 ± 0.93	86.58 ± 3.25	81.76 ± 1.92	79.13 ± 1.65
	MCD	91.86 ± 1.88	90.33 ± 1.06	87.56 ± 2.02	82.54 ± 2.15	78.34 ± 2.31
	DDC-LMMD	94.35 ± 1.06	95.12 ± 1.78	94.23 ± 1.89	92.66 ± 2.13	91.56 ± 2.24
Source Combine	DAN	95.12 ± 0.14	94.13 ± 0.33	93.56 ± 0.43	93.44 ± 0.47	93.21 ± 0.62
	DANN	95.53 ± 0.13	94.08 ± 0.45	93.06 ± 0.12	92.45 ± 0.39	91.19 ± 0.93
	DDC-LMMD	96.76 ± 0.09	96.35 ± 0.16	95.32 ± 0.35	95.13 ± 0.41	94.51 ± 0.44
Multiple Sources	ADACL	96.81 ± 0.32	95.47 ± 0.38	93.70 ± 0.42	92.71 ± 0.81	92.36 ± 1.16
	MADN	96.32 ± 0.12	95.67 ± 0.21	93.20 ± 0.47	92.47 ± 0.65	91.67 ± 1.23
	MSSA	99.52±0.09	99.48±0.09	99.18±0.10	98.97±0.17	98.30±0.38

under imbalanced target domain training samples, an imbalanced target domain training set is generated on the CWRU dataset, and the detailed scheme is shown in Table 14. The mini-batch size was set to 64 for all comparison methods.

The diagnostic results of the different methods are presented in Fig. 21 and Table 15. The proposed MSSA still achieved a high diagnostic performance under an unbalanced training set, with an accuracy of 97.35%, which is significantly superior to other diagnostic methods. Except for DAN-LMMD, other SUDA diagnostic methods had poor diagnostic effects in this experiment. This may be because global domain adaptation is more inclined to classify samples into normal categories, and local domain adaptation considers the alignment of the distribution of each subdomain, which can resist the bias of sample weight allocation caused by the imbalanced training set. The diagnostic performance of the multi-source domain fault diagnosis method under an imbalanced training set is significantly better than that of the single-source domain method, in which ADACL and MADN have relatively high diagnostic accuracy. This indicates that using multi-classifier alignment or multi-domain identifier adversarial training can help reduce the bias caused by an imbalanced training set. The proposed method matches the fine-grained distribution of relevant sub-domains in each pair of source and target domains; therefore, the classification of different categories is rarely affected by imbalanced training samples. The stability and sensitivity of the proposed model for the recognition of each state category were further analyzed. Fig. 22 illustrates the receiver operating characteristic (ROC) curve for each diagnostic task. It can be clearly seen that in different diagnostic tasks, the area under curve (AUC) of most categories of the proposed method is equal to 1, and even in task A, C, D→B, the smallest AUC reaches 0.9755. This indicates that the proposed model can effectively identify each fault state and has excellent stability even under imbalanced training sample conditions.

Table 14

Design of training dataset in the target domain.

Condition label	0	1	2	3	4	5	6	7	8	9
Proportion of samples	100%	25%	50%	25%	25%	50%	25%	25%	50%	25%

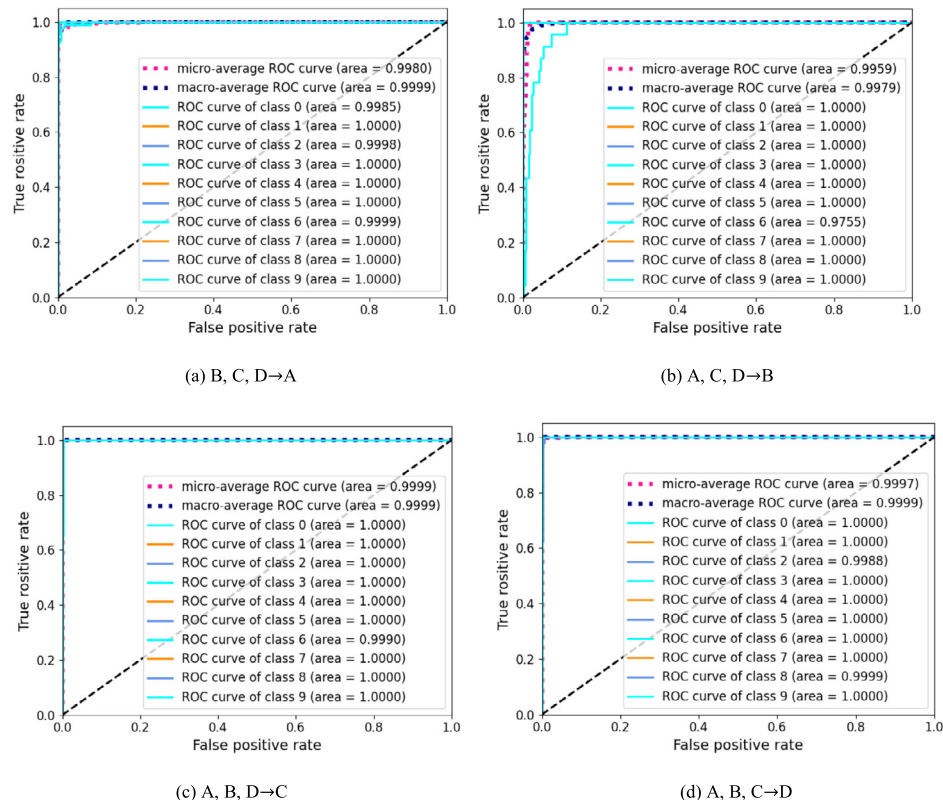
5. Conclusion

Most existing UDA methods consider only the transfer of information from a single source domain to the target domain. In view of the insufficient transfer information of the single source domain, a novel multi-source information transfer learning framework with subdomain adaptation was developed in this study, for the fault diagnosis of rotating bearings. The network model simultaneously aligns the feature space distribution of each pair of source and target domains through multiple branch structures to learn multiple source domain-invariant representations. In addition, LMMD was applied to achieve precise alignment of the distribution of related subdomains of the same category. Finally, multiple source classifiers were combined with the corresponding weighted score to perform a joint diagnosis of the equipment status. The effectiveness of this method was verified using three rotating machinery datasets. Experimental analysis shows that MSSA surpasses other single-source and global domain adaptation approaches in terms of diagnostic performance, and even when the target domain training samples are few or imbalanced, it still achieves a high diagnostic accuracy. Although the proposed method achieved a good diagnostic effect in this study, it is usually only the target domain training data of the health status that can be collected in more practical industrial production. The use of the source domain data and health status target domain data to establish an effective generalization model is a problem worthy of study.

Table 15

Average diagnostic accuracy and standard deviation of different methods on the CWRU dataset with imbalanced training sample in the target domain.

Standards	Method	B,C,D→A	A,C,D→B	A,B,D→C	A,B,C→D	Average
Single Best	DAN	78.72 ± 1.16	75.62 ± 2.23	81.34 ± 1.05	80.61 ± 2.12	79.07
	JAN	83.26 ± 1.31	78.26 ± 1.64	85.16 ± 2.16	86.58 ± 1.86	83.32
	DANN	79.22 ± 2.87	75.77 ± 2.14	81.69 ± 1.32	82.88 ± 1.06	79.85
	MCD	82.36 ± 2.21	77.65 ± 1.33	83.69 ± 2.16	84.72 ± 1.61	82.11
Source Combine	DDC-LMMD	90.94 ± 1.37	89.33 ± 1.52	91.67 ± 0.81	90.39 ± 1.19	90.58
	DAN	93.33 ± 0.93	91.39 ± 1.12	92.50 ± 1.35	91.67 ± 1.88	92.22
	DANN	92.22 ± 1.36	90.17 ± 1.21	93.28 ± 1.62	93.56 ± 1.63	92.31
	DDC-LMMD	93.28 ± 1.03	92.17 ± 0.62	93.56 ± 0.62	94.00 ± 0.94	93.25
Multiple Sources	ADACL	95.39 ± 0.89	93.72 ± 1.33	94.35 ± 0.67	95.38 ± 1.06	94.77
	MADN	95.45 ± 1.03	93.22 ± 0.94	94.78 ± 0.73	95.17 ± 0.86	94.66
	MSSA	97.61±0.67	95.53±0.79	98.72±0.42	98.06±0.39	97.35

**Fig. 22.** Receiver operating characteristic curve.

CRedit authorship contribution statement

Jinghui Tian: Conceptualization, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Dongying Han:** Supervision, Project administration, Funding acquisition. **Mengdi Li:** Writing – review & editing. **Peiming Shi:** Methodology, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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